

Solución (01)

Usamos:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt} \quad [1]$$

$$\Phi_B = BA \quad [2]$$

[2] EN [1]

$$\mathcal{E} = -N \frac{dBA}{dt}$$

$$\mathcal{E} = -NA \frac{dB}{dt} \quad [3]$$

DEL PROBLEMA

$$B(t) = B_0 e^{-t/\tau} \quad [4]$$

$$\mathcal{E} = -NA \frac{d(B_0 e^{-t/\tau})}{dt}$$

$$\mathcal{E} = -NAB_0 e^{-t/\tau} \frac{d(-t/\tau)}{dt}$$

$$\mathcal{E} = -NAB_0 e^{-t/\tau} \left(-\frac{1}{\tau}\right) \frac{dt}{dt}$$

$$\mathcal{E} = \frac{NAB_0}{\tau} e^{-t/\tau} \quad [5]$$

USAREMOS

$$P = I^2 R \quad [6]$$

$$\mathcal{E} = IR \quad [7]$$

$$I = \frac{\mathcal{E}}{R} \quad [8]$$

[8] EN [6]

$$P = \frac{\mathcal{E}^2}{R^2} R$$

$$P = \frac{\mathcal{E}^2}{R} \quad [9]$$

[5] EN [9]

$$P = \left[\frac{N^2 A^2 B_0^2}{R \tau^2} \right] e^{-2t/\tau} \quad [10]$$

LA ENERGÍA DISIPADA EN LA RESISTENCIA ESTÁ DADA

$$E = \int P dt \quad [11]$$

[10] EN [11]

$$E = \int_0^t \left[\frac{N^2 A^2 B_0^2}{R \tau^2} \right] e^{-2t/\tau} dt$$

RESOLVIENDO

$$E = \left[\frac{N^2 A^2 B_0^2}{R \tau} \right] \int_0^t e^{-2t/\tau} d\left(\frac{t}{\tau}\right)$$

$$E = \left[\frac{N^2 A^2 B_0^2}{R \tau} \right] \left(\frac{1}{-2} \right) \int_0^t e^{-\frac{2t}{\tau}} d\left(-\frac{2t}{\tau}\right)$$

$$E = \left[\frac{N^2 A^2 B_0^2}{R\tau} \right] \left(-\frac{1}{2} \right) \int_0^t d e^{-2t/\tau}$$

$$E = \left[\frac{N^2 A^2 B_0^2}{R\tau} \right] \left(-\frac{1}{2} \right) \left[e^{-\frac{2t}{\tau}} \right]_0^t$$

$$E = \left[\frac{N^2 A^2 B_0^2}{R\tau} \right] \left(-\frac{1}{2} \right) \left(e^{-\frac{2t}{\tau}} - e^0 \right)$$

$$E = \left[\frac{N^2 A^2 B_0^2}{2R\tau} \right] \left(1 - e^{-\frac{2t}{\tau}} \right)$$

$$E = \frac{(NA B_0)^2}{2R\tau} \left(1 - e^{-\frac{2t}{\tau}} \right)$$

REEMPLAZANDO VALORES

$$E = \frac{(18\pi(0,100\text{m})^2(0,50\text{T}))^2}{2(2,0\Omega)(0,10\text{s})} \left(1 - e^{-2t/0,50\text{s}} \right)$$

$$E = 0,20 \left(1 - e^{-20t} \right) \text{ J}$$

Solução (02)

$$M = \frac{\Phi_{12}}{I_1}$$

$$M = \frac{1}{I_1} \int_{l_1}^{l_2} \frac{\mu_0 I_1 \omega}{2\pi r} dr$$

$$M = \frac{1}{\cancel{I_1}} \left(\frac{\mu_0 \cancel{I_1} \omega}{2\pi} \right) \int_{l_1}^{l_2} \frac{dr}{r}$$

$$M = \frac{\mu_0 \omega}{2\pi} \int_{l_1}^{l_2} d(\ln r)$$

$$M = \frac{\mu_0 \omega}{2\pi} (\ln r) \Big|_{l_1}^{l_2}$$

$$M = \frac{\mu_0 \omega}{2\pi} [\ln l_2 - \ln l_1]$$

$$\boxed{M = \frac{\mu_0 \omega}{2\pi} \ln \left(\frac{l_2}{l_1} \right)}$$

b) Usamos

$$Q = Q_0 [\cos(\omega t + \phi)]$$

De a) tenemos

$$Q = \frac{\sqrt{2}}{2} Q_0$$

$$\phi = 0$$

Obtenemos

$$\frac{\sqrt{2}}{2} Q_0 = Q_0 \cos(\omega t + 0)$$

$$\cos \omega t = \frac{\sqrt{2}}{2}$$

Trigonometría

$$\omega t = \frac{\pi}{4}$$

$$\omega = \frac{2\pi}{T}$$

$$\frac{2\pi}{T} t = \frac{\pi}{4}$$

$$8t = T$$

$$t = \frac{T}{8}$$

04) 1) DETERMINANDO LA FRECUENCIA

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{30 \cdot 10^{-3} \cdot 12 \cdot 10^{-8}}}$$

$$\omega_0 = \frac{1}{\sqrt{36 \times 10^{-8}}}$$

$$\omega_0 = \frac{1}{6 \times 10^{-4}}$$

$$\omega_0 = \frac{10^4}{6}$$

$$2\pi f_0 = \frac{10^4}{6}$$

$$f_0 = \frac{10000}{12\pi}$$

$$f_0 = 265,257 \text{ Hz}$$

$$f_0 = 265,3 \text{ Hz}$$

$$f_0 = 265 \text{ Hz}$$

Solución (03)

Usamos

$$U = \frac{1}{2} \left(\frac{Q^2}{C} \right)$$

2 >

$$U = U_0$$

$$\frac{Q^2}{2C} = \frac{1}{2} \left(\frac{Q_0^2}{2C} \right)$$

$$\frac{Q^2}{C} = \frac{Q_0^2}{2C}$$

$$Q^2 = \frac{Q_0^2}{2}$$

$$Q = \frac{Q_0}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}}$$

$$\boxed{Q = \frac{\sqrt{2}}{2} Q_0}$$

$$\phi = \arctan \left(\frac{94,248 - 26,52}{25} \right)$$

$$\phi = \arctan (2,71)$$

$$\phi = 69,75^\circ$$

$$\phi = 69,8^\circ$$

$$\phi = 70^\circ$$

1) DETERMINA ϕ

$$\phi = \arctan \frac{X_L - X_C}{R}$$

Si

$$X_L = \omega L$$

$$X_L = 2\pi f L$$

$$X_L = 2\pi 500 \cdot 30 \cdot 10^{-3}$$

$$X_L = 30000 \times 10^{-3} \pi$$

$$\left. \begin{array}{l} X_L = 30\pi \Omega \end{array} \right\} \left. \begin{array}{l} X_L = 94,248 \Omega \end{array} \right\}$$

Ademas

$$X_C = \frac{1}{\omega C}$$

$$X_C = \frac{1}{2\pi f C}$$

$$X_C = \frac{1}{2\pi \cdot 500 \cdot 12 \cdot 10^{-6}}$$

$$X_C = \frac{1}{12000 \times 10^{-6} \pi}$$

$$\left. \begin{array}{l} X_C = 26,52 \end{array} \right\}$$

$$X_C = \frac{1000}{12 \cdot 3}$$

$$\left. \begin{array}{l} X_C = \frac{250}{3\pi} \Omega \end{array} \right\} \left. \begin{array}{l} X_C = \frac{83,33}{\pi} \Omega \end{array} \right\}$$

c) El campo magnético producido por una espira circular o un anillo de corriente es:

$$d\vec{B}_{\text{anillo}} = \frac{\mu_0 dI r^2}{2(r^2+z^2)^{3/2}}$$

$$\vec{B}_{\text{anillo}} = \frac{\mu_0 \sigma \omega r^3 dr}{2(r^2+z^2)^{3/2}}$$

$$\vec{B}_{\text{disco}} = \frac{\mu_0 \sigma \omega}{2} \int_0^R \frac{r^3 dr}{(r^2+z^2)^{3/2}} = \int_0^R \left(\frac{\mu_0 \sigma \omega}{4} \right) \frac{r^2 dr}{(r^2+z^2)^{3/2}}$$

Si $t = (r^2+z^2)^{3/2}$
 $\rightarrow t^{2/3} = (r^2+z^2) \rightarrow \frac{r^2}{2} = \frac{t^{2/3} - z^2}{2} \rightarrow \frac{r^2}{2} = \frac{t^{2/3} - z^2}{2}$

Entonces: $\vec{B}_{\text{disco}} = \frac{\mu_0 \sigma \omega}{4} \int_{t_0}^{t_f} \frac{(t^{2/3} - z^2)^{3/2} dt}{t}$

$$\vec{B}_{\text{disco}} = \frac{\mu_0 \sigma \omega}{6} \int_{t_0}^{t_f} \frac{(t^{2/3} - z^2)^{3/2}}{t} dt = \frac{\mu_0 \sigma \omega}{6} \left[\int_{t_0}^{t_f} t^{-1/3} z^2 dt \right]$$

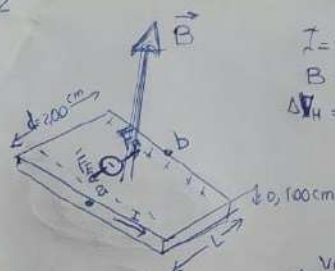
$$\vec{B}_{\text{disco}} = \frac{1}{6} \mu_0 \sigma \omega \left[3 t^{1/3} z^2 (-3) t^{-1/3} \right]_{t_0}^{t_f} = \frac{1}{2} \mu_0 \sigma \omega \left[t^{1/3} + z^2 t^{-1/3} \right]_{t_0}^{t_f}$$

$$\vec{B}_{\text{disco}} = \frac{1}{2} \mu_0 \sigma \omega \left[\sqrt{r^2+z^2} + z^2 (r^2+z^2)^{-1/2} \right]_0^R$$

$$\vec{B}_{\text{disco}} = \frac{1}{2} \mu_0 \sigma \omega \left[\sqrt{r^2+z^2} + \frac{z^2}{\sqrt{r^2+z^2}} \right]_0^R$$

$$\vec{B}_{\text{disco}} = \frac{1}{2} \mu_0 \sigma \omega \left[\sqrt{R^2+z^2} + \frac{z^2}{\sqrt{R^2+z^2}} - 2z \right]$$

2



$I = 20,0 \text{ A}$
 $B = 2,00 \text{ T}$
 $\Delta V_H = 4,27 \mu\text{V}$ Hallar velocidad de los electrones

2) $\vec{F}_E + \vec{F}_B = 0$

$\rightarrow F_E = F_B$

$q E_H = q v_d B$; pero $\Delta V_H = E_H L$

Velocidad de portadores de carga $\rightarrow v_d = \frac{E_H}{B} = \frac{\Delta V_H}{L B} = \frac{4,27 \cdot 10^{-6}}{(2 \cdot 10^{-2} \cdot 2,00)}$

$v_d = 1,07 \cdot 10^{-4} \text{ m/s}$

b) Halla la densidad de los portadores de carga

$J = n q v_d = \frac{I}{A_T}$

$\rightarrow n = \frac{I}{q A v_d} = \frac{20,0}{(1,6 \cdot 10^{-19})(2,00 \cdot 10^{-2})(1,07 \cdot 10^{-4})}$

$\Rightarrow n = 5,84 \cdot 10^{23} \text{ portadores/m}^3$